

Thinking Critically about the Bioeconomy



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Pedagogical Principles

- Economics and Ecology as Democratic
- Diversity and Plurality
- Embodied practice
- Reforesting our Shared Imaginations



Class Outlines

- Thinking Critically through Bioeconomics, May 14th
- Plural Perspectives on Economy and Ecology, May 15th
- Solidarity and feminist approaches to ecological economics, May 19th
- Alternative Policies and Imagining the Future, May 21st
- Seminar, May 28th

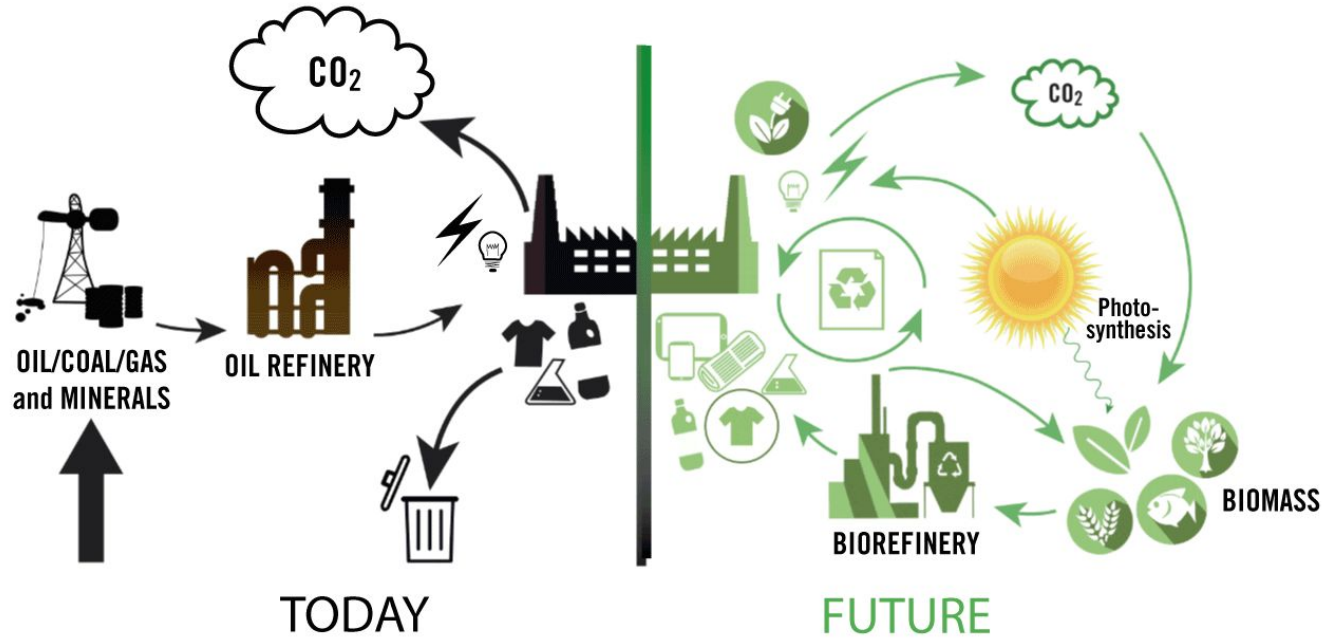
Core Questions

Why is it important to study and think critically about the bioeconomy today?

How is the bioeconomy relevant to your own life?

Why





Why

From resource economy to bioeconomy

An important part of Sweden's economic transition strategy – apart from reducing emissions – is about actively trying to use natural processes to produce energy, industrial products and much else. This so-called bioeconomy involves much more than making things more environmentally friendly. Sweden is pioneering ways to use natural materials that are 100 per cent recyclable and can be part of the 'cradle-to-grave' process.

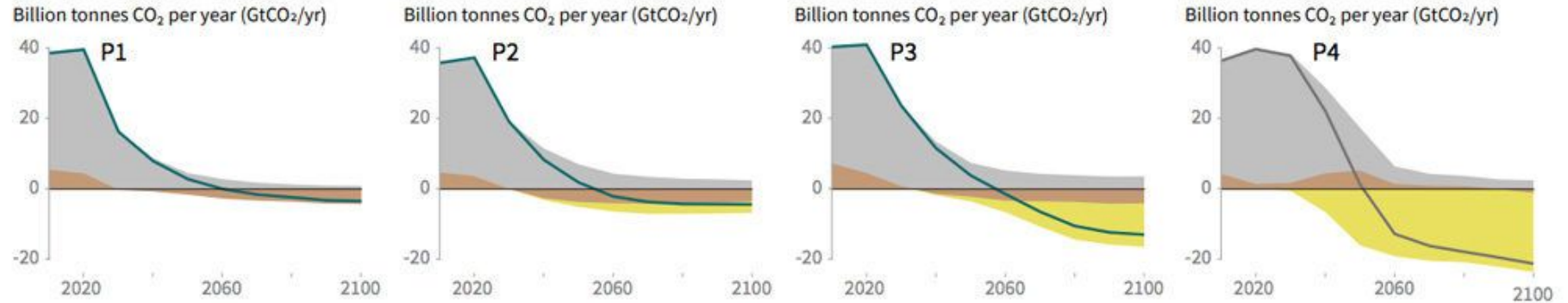
Sweden has a wealth of sustainable natural resources to work with. Already, most of the energy consumed comes from renewables, and its managed forests already provide the main supply of wood products to the EU.

Source: Swedish government website, retrieved May 2025.

Why

Breakdown of contributions to global net CO₂ emissions in four illustrative model pathways

● Fossil fuel and industry ● AFOLU ● BECCS





*Karl Polanyi
&
Ilona Duczynska*

Polanyian ideas

- Substantivism (economics is embedded in culture-society)
- *The Great Transformation*: a market society emerges through transformation of collective mentalities, and ripping of social fabric
- The idea of self-regulating markets is fantasy & fiction
- Economics is constituted by politics & social decisions
- Double movement (marketization vs. social protection)
- Towards an economics in service to society

Karl Polanyi

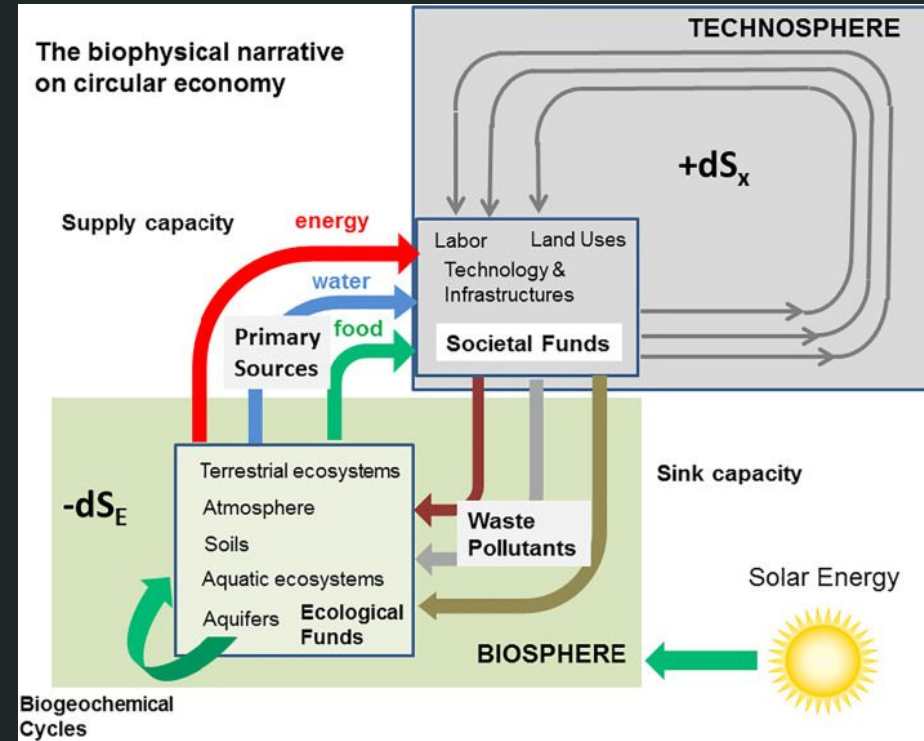
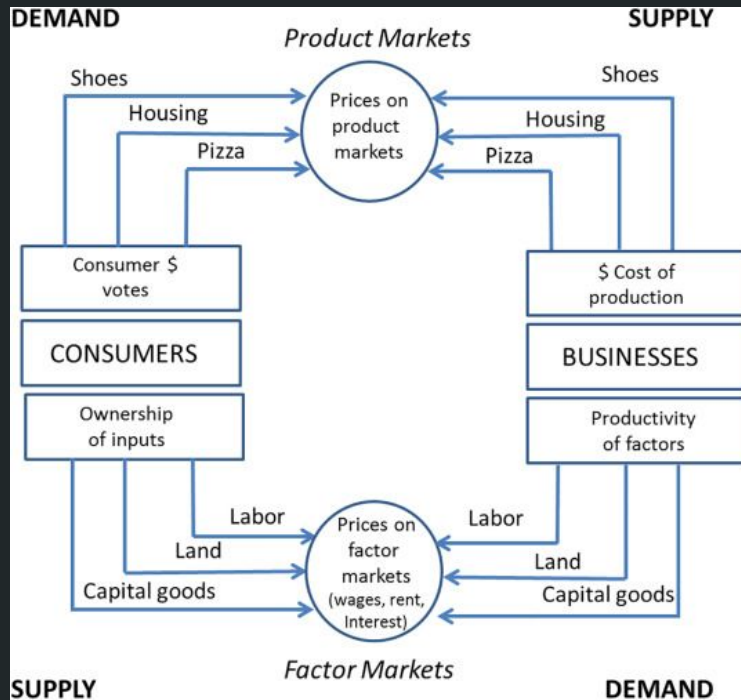
“The outstanding discovery of recent historical and anthropological research is that man’s economy, as a rule, is submerged in his social relationships.” - *The Great Transformation*

Bioeconomy - a history

- **1920s** - fisheries scientist Fedor Baranov talks of bioeconomics, reflecting on importance of understanding ecological limits of ecosystems
- **1970s** - bioeconomy & bioeconomics proposed by Georgescu-Roegen (Vivien et al., 2019), as vision of economics in touch with physical & material limits to growth, and regeneration rhythms of ecologies
- **Early 2000s** - bioeconomy emerges from literature on biotechnology and sustainable development, to become a buzzword on international scene.
- **2006** - OECD launches 'Bioeconomy to 2030' framework.
- **2012** - EU launches Bioeconomy Strategy, largely “technological innovation as a means to increase the use of bio-based resources in industrial processes”.

Diverse visions of the bioeconomy

- **Biotechnology, bioresource, and bio-ecology** (Stockholm Environment Institute).
- For (Giampetro 2019), there are two broadly competing bioeconomy visions: a) an **economic paradigm** based on technological progress and perpetual growth; an **entropic-thermodynamic perspective** that reflects on limits to growth.
- **Plantation systems or agro-biodiverse systems** (Ossi I. Ollinaho & Kröger 2023)
- **Bioeconomics, or socio-bioeconomics**



“the narratives of EU and bioeconomy as presented in (global) politics and by important interest groups have a theoretical basis in neoclassic models that endorse a strategy of top-down planning of technological fixes typical of the neoliberal ideology” - (Giampetro, 2019)

Socio-economic dynamics surrounding the bioeconomy

- **Disproportionate flows:** ‘a lock-in of genetic engineering and related ‘new biotechs’, alongside a lock-out of agroecology’ (Levidow 2019)
- **Corporate appropriation:** ‘Much of the literature suggests that the EU bioeconomy policy has been appropriated by powerful corporate and industry interests, and governmental actors in support of economic growth and the industrial use of natural resources’ (Suominen et al. 2022)



“In the case of Brazil nuts, for example, 79% of the profits go to the processing industry and only 11% remain with the forest communities. Regarding cocoa, rural producers and agro-extractives in Brazil get up to 6% of the money, while processing companies based in Europe get 24% of the profits. Large international chocolate producers, like Mars Inc. and Nestlé, own more than 70% of the profits.”

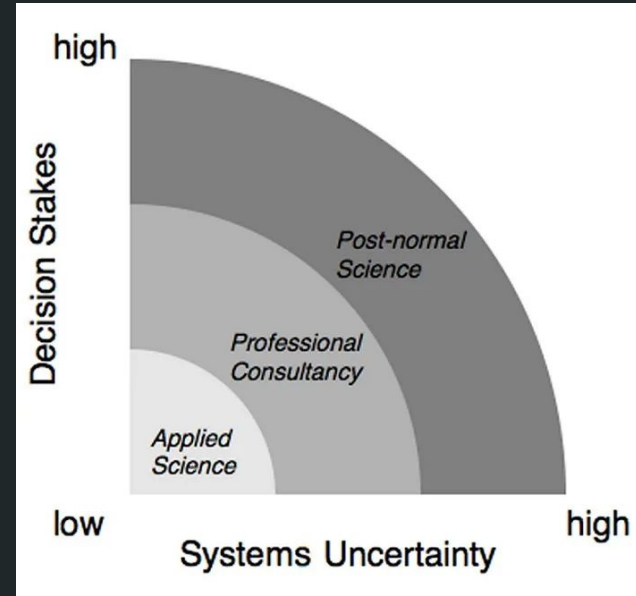
Forms of Power in Transitions

- “(a) innovative power, or the capacity to create new resources (e.g., technical innovations);
- (b) transformative power, or the ability to change institutional settings (changing policies, obtaining new incentives, etc.);
- (c) reinforcing power, the capacity to reproduce—and, eventually, reinforce—existing institutions”

(Bastos Lima 2021; Avelino 2017)

An Ontological Crisis in Sustainability Science

- Giampetro 2023: ‘the circular bioeconomy concept epitomizes the ontological crisis currently experienced by sustainability science’
- For Giampetro, the bioeconomy lacks agreed definition, in science and politics, which together with different ontological “takes on what the bioeconomy is and isn’t, leads to a “a systemic lack of quality control on the science–policy interface.”



Traditional Ecological Knowledge Systems & the Bioeconomy



Sources: Amazon Frontlines / Alianza Ceibo



Non-hegemonic visions of the bioeconomy



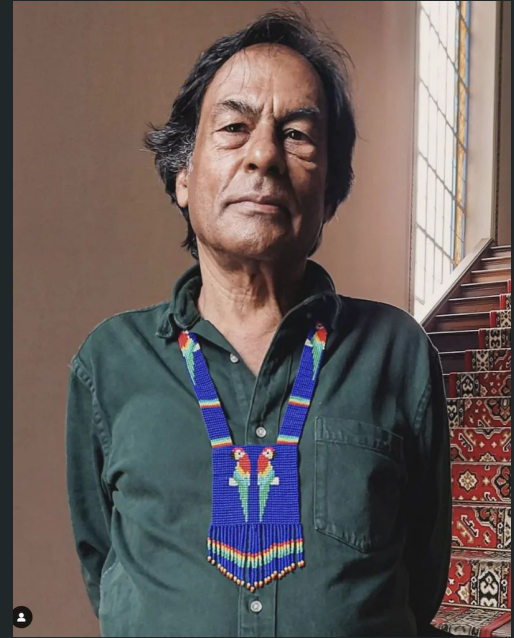
“We had never used the word bioeconomy until very recently...For us, this is ancestral knowledge, something we’ve been doing for generations.” - Telma Taurepang, general coordinator of the Union of Indigenous Women of the Brazilian Amazon (UMIAB)

Non-hegemonic visions of the bioeconomy

“Those immersed in the market economy only know how to think monoculturally.”

“We must reforest our imagination - and maybe, just maybe, we can reconnect with a poetics of urbanity that restores life’s vitality, instead of simply repeating Greeks and Romans.”

“In the so-called environmental sphere, the discourse takes a different form, namely by creating reserves. The calculation of ecological sustainability is the reserve. That is nothing other than an economic discourse: save something today in order to eat it tomorrow. This is a viewpoint that focuses solely on the needs of humans and couldn’t care less about any other living creatures. Whales, elephants, birds, forests ... This entire discourse serves the sole purpose of control. In my opinion, nature conservation is control!” - Ailton Krenak



“The future is ancestral and humanity needs
to learn with it to step softly on the earth.”

— Ailton Krenak, Indigenous educator and philosopher

Non-hegemonic visions of the bioeconomy



“What the white people call “nature’s protection” is actually us, the forest people, those who have lived under the cover of its trees since the beginning of time...” - Davi Kopenawa

Non-hegemonic visions of the bioeconomy



“But for Indigenous peoples it is clear: the less you know about something, the less value it has to you, and the easier it is to destroy. And by easy, I mean: guiltlessly, remorselessly, foolishly, even righteously. And this is exactly what you are doing to us as Indigenous peoples, to our rainforest territories, and ultimately to our planet’s climate.”

You forced your civilisation upon us and now look where we are: global pandemic, climate crisis, species extinction and, driving it all, widespread spiritual poverty. In all these years of taking, taking, taking from our lands, you have not had the courage, or the curiosity, or the respect to get to know us. To understand how we see, and think, and feel, and what we know about life on this Earth.” - Nemonte Nenquimo

Bioeconomy from below



[Who we are](#) ▾

[What we do](#) ▾

[How we do it](#) ▾

[Get involved](#) ▾



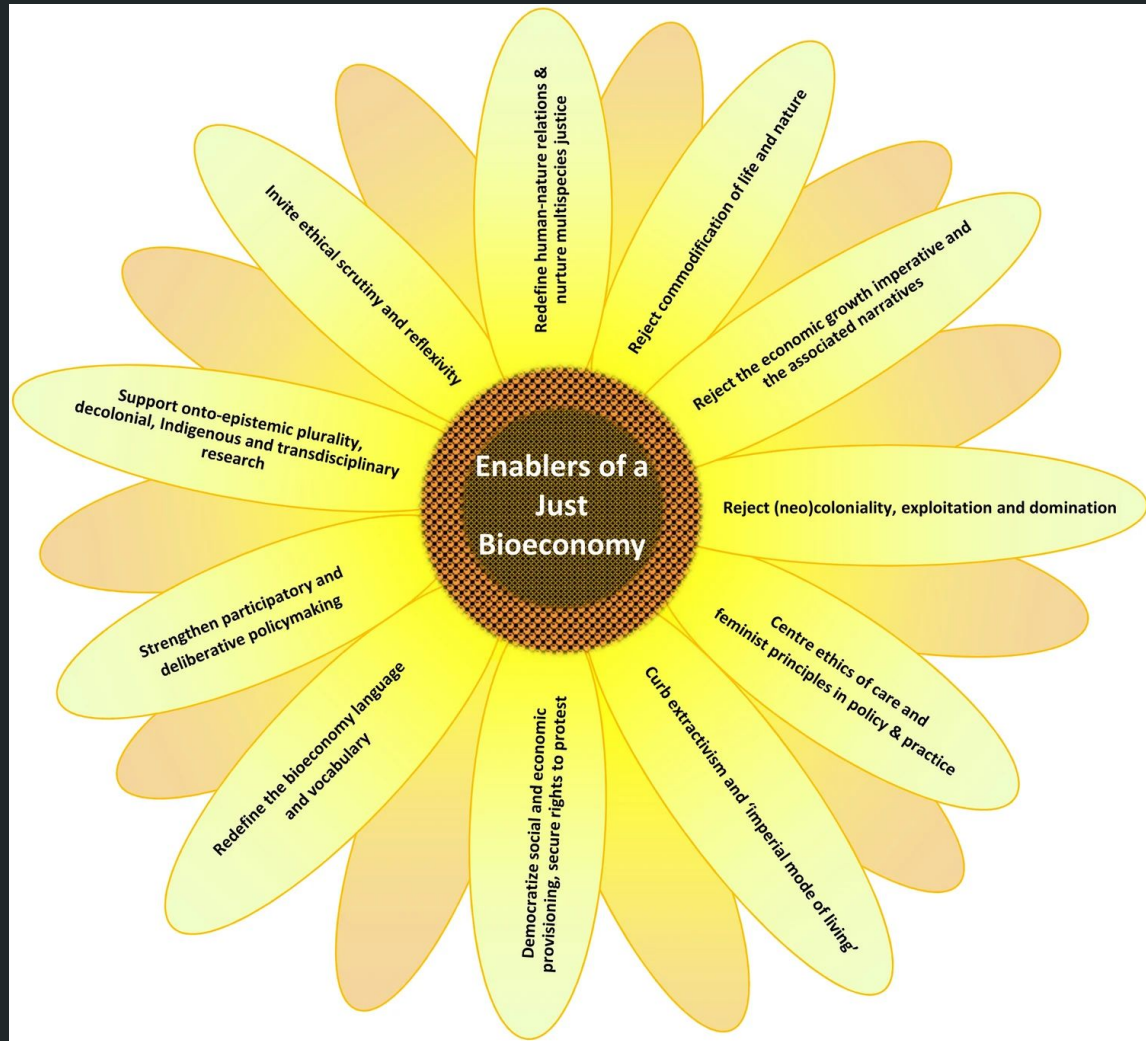
Empowering Communities for Future.

ECOLISE is the European Network for Community-Led Initiatives on Climate Change and Sustainability



A relational approach to the bioeconomy

(Ramcilovic-Suominen et al. 2025)



Care bioeconomy

“From a (re)productivity perspective, sustainability is facilitated by an orientation toward the three principles of a caring economy (precaution, cooperation, and orientation toward what is necessary for life), which replace the dominant principles of aftercare, competition, and orientation toward monetary measures. Importantly, the third strategic principle points to sufficiency as a sustainability strategy, which differs from current emphases on eco-efficiency and technological optimism within the bioeconomy and unlocks new routes for the sustainable organization of society.” -

Additional ideas

The economy as political performance



Tools for thinking through greenwashing

MSCI, the largest ESG rating company, doesn't even try to measure the impact of a corporation on the world. It's all about whether the world might mess with the bottom line.

By [Cam Simpson](#), [Akshat Rathj](#), and [Saijel Kishan](#)

Environmental factors ● are the most deceptive to the uninitiated, because MSCI rates the potential impact of the world on the company, not the company's impact on the world. For example, “**Water Stress**” ● measures whether the local community has enough water for the company, not whether the company is stressing the local water supply. In total, 32% of the upgrades with an environmental factor included this.

Source: Bloomberg, 2021

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— Ailton Krenak, Indigenous educator and philosopher